Digital Design:-

- Conversion of one number system into another.
- Question on one's and 2’s complement and XS-3 code,Binary to Gray converter and vice versa.
- Why NAND and NOR are called as universal gates and implement each one of them
- A function is given in either pos or sop and you can be asked to impllement in either NAND or NOR.
- Questions on Full adder and full subtractor and Look ahead carry adder(Many questions could be asked so it's better to understood the concepts)
- A Boolean function is given and you could be asked to minimize it using either K-map or Tabulation method.
- Conceptually understood all the laws and Theorems related to Digital logic
- Implement all the basic gates using 2:1 Mux
- A function in the form like f(a,b,c) = sum(1,3,4,6) is given and can ask you to implement using 8:1 Mux and 4:1 Mux.(You need to muster the concept of Mux tree very well as many twisted question can be asked)
- Implement 4:1Mux using 2:1 Mux without using any additional gates.
- Implement a full adder using two 4:1 Mux
- Design a 16:1 Mux using 2:1 Mux
- Implement a 2:1 Mux using Tristate buffers
- Implement the following function using 2:1 Mux
- Design a full adder using 3:8 Decoder(Both active high and low)
- Questions on priority encoder and it's implementation.
- Difference between latch and flip flop
- Conversion of one flip to another like JK to SR,T to D etc
- SISO and PIPO implementation
- Different number of cycles required to implement Johnson Counter,Ring Counter,Ripple counter.
- Decade counter and up/down counter implementation
- Mod counter with specified duty cycle implementation.
- Sequence Detector like 10101 is given and you are asked to implement this FSM depending on overlapping and non overlapping method(High Priority)
- Difference between Mealy and Moore.
- Different types of Hazards and their usage.
- Difference between Setup time and Hold time and explain with waveform.
- Propagation delay and contamination delay.
- Questions on Clock skew,Slack and Slew.
- Hold slack calculation

    - A given circuit is given and you could be asked to find the frequency of that.
    - Divide by 2 counter implementation
- Complete understanding of the ASIC and the FPGA flow.
- What are CLB,IOB and LUT,s in FPGA
- Asynchronous and Synchronous FIFO with special focus on FIFO depth calculation
- Reset strategies and Reset Recovery time.
- Memory Controller and it's implementation in Verilog.
    - Different gates using Mux
    - gray code application
    - Tri-state buffer
    - depth of fifo
    - fan-in vs fan-out
    - Clock-gating
    - What are the different types of skews used in VLSI
    - Explain what is Slack?
- **Design a clock divider circuit which divides the clock by an odd number and has 50% duty cycle. (o/p clk = i/p clk/N, where N is an odd number).**
    - What is meant by race condition
    - What is one hot encoding
    - Explain the difference between binary and gray encoding and the benefits of each?
    - universal gates
- Implement 2:1 Mux using Tristate buffers.
- Design 4:16 Decoder using 3:8 Decoder.
- ***Design Full adder using 3:8 Decoder.***
- Implement a counter with Mux
- ***Design a MOD 10 counter with 50%/33% duty cycle.***
    - What are ***synchonisers*** and when to use them
    - Between Binary and ***Gray Counter***,which one to use and why?
    - For an input clk of f frequency generate an output of 2*f frequency
    - Output should be high for only odd +ve edge transition
    - No. of 2*4 decoders needed to implement 4*16 Decoder
    - what is synchronous and asynchronous counter?
    - XOR vs XNOR –special characteristics.
    - Edge detection circuitry.
- Noise In digital Cirguits.

- Noise Margin.

Verilog/VHDL:-

- Difference between Blocking and Nonblocking statements.
- Difference between Intra and Inter assignment delay
- Difference between Task and Function.
- Difference between reg and wire
- A code could be given on blocking and Nonblocking and you might be asked to provide the output.
- Difference between Transport and Inertial delay.
- Different types of wait statements in VHDL and explain each of them.
- Write a verilog code for asynchronous and Synchronous D flip flop
- Why latches are not inferred in RTL Design
- Different Coding guidelines for RTL Design(Refer Sunburst Design Papers)
- Difference between Verilog Full case and Parallel case
- Can a task called a function?If so then how it is possible.
- Verilog code to swap the content of register,with and without using a temporary variable.
- Difference between $monitor and $strobe.
- Difference between Verilog and VHDL.
- What does **if else** statements synthesized to and what does **case** statements synthesized into?
- Difference between Case equality and case inequality.
- Explain the stratified Event Queue in Verilog.
- Difference between signal and variable in VHDL.
- What is Delta Delay in VHDL?

- Different kinds of modelling technique in VHDL and explain each one of them
- **In Verilog code what does “timescale 1 ns/ 1 ps” signifies?**
- **Explain what is the use of defpararm?**
- **How Verilog is different to normal programming language?**
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